

Taiwan Cooling Components

Accelerating cooling upgrades in Nvidia VRU and ASICs to unlock continued TDP growth

We see an accelerating cooling spec upgrade trend in both AI ASICs and GPU systems, and the content uplift looks larger than investors' expectations, especially for Nvidia's next-gen VR Ultra (VRU), contrary to recent market concerns on potential de-specing. We continue to believe that cooling is the key bottleneck to unlocking continued TDP growth of all AI chips and migration of AI server systems. For VRU, we see new designs and spec upgrades across **cold plate (double-sided design and diamond copper material adoption), quick disconnect (MQD upgrading to ZQD), and lid/heat spreader (3-piece/removable lid design)**. We believe the stronger-than-expected content growth in Nvidia VRU, along with the further increasing ASIC and AMD GPU cooling value will continue to benefit the leading suppliers of each component. Looking forward to the next 1-2 years, we believe the CPO/NPO switches and higher-spec optical transceivers/modules will become the next driver with mainstream adoption of liquid cooling in future generations (likely from 1.6T/3.2T gens). We prefer AVC (OW) and Fositek (OW) in the cooling space.

- **Cold plate: double-sided design likely adopted in Nvidia VRU, AWS Trn3 and AMD MI450 to drive +50% content; diamond copper to add value.** The double-sided cold plate design looks set to become the next step-up in cooling specs. We expect adoption in AWS Trainium 3 (Trn3) and AMD MI450 compute trays, with Nvidia VRU likely next. This new double-sided design was required given the VPD (Vertical Power Delivery) architecture, which moves the power discretes, SOCAMM, and capacitors to the backside of chip board, requiring incremental cold plates for backside cooling. VRU Bianca cold plates may also upgrade their material to "diamond copper" for better cooling effectiveness, which adds further value per unit. This combination supports meaningful cooling content upside for VRU (~50% increase vs. VR200) versus market expectations for flat-to-down cooling content on de-spec concerns. Outside these platforms, Google TPU v8 should drive solid cooling content via gold-plated cold plates plus additional units adopted for surrounding ICs. TPU v9's larger chip die and higher TDP also increase the odds of the double-sided cold plate adoption.
- **Quick disconnect (QD): No content change with the potential VRU 9-switch tray design of 1.5U with "double-layer architecture"; meaningful upgrades in VRU (ZQD adoption) with ~100% content increase per rack.** In the VRU system, we believe Nvidia will upgrade some of its QDs from the previous MQD (Multi Quick Disconnect)/UQD (Universal Quick Disconnect) to the higher-spec ZQD (Zero-drip/flow rate Quick Disconnect) with ~2x ASP, in order to help improve the cooling efficiency with better liquid flow rate and limited leakage. Per JPMe, the total QD content per VRU NVL72 rack will increase by ~100% vs VR200/GB300 racks, as the ZQD with 2x ASP will be adopted in 1) the compute tray Bianca modules (upgrading to ZQD06 spec along with GPU cold plate upgrades), 2) entry QDs for all 9 compute trays and 9 "double-layered" switch trays (the 2 out-of-tray plug QDs upgrading to ZQD14 spec), and 3) rack manifold socket QDs (upgrading to ZQD14 as well).
- **Heat spreader: No delay of 3-piece/removable lid adoption in Rubin Ultra, even pulling forward some Rubin GPU potentially from 4Q26/1Q27.** We believe Nvidia's Rubin Ultra GPU will adopt the "removable lid + stiffener +

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multiple screws” design (or the so-called 3-piece lid design) with no delay, which should drive the heat spreader/lid content increase by 4-5x vs the current 1-piece lid design for Rubin GPU. Based on our industry check, Nvidia might even adopt this new design in small volume of Rubin GPU potentially from 4Q26 for trial production and enter mass production in the next gen Rubin Ultra GPU potentially in 1Q27. We believe the adoption schedule is slightly ahead of market expectations, which might provide near-term rev/earnings upside for the 3-piece lid supplier. In addition, AMD MI450 series adoption of 2-piece lid (~10x ASP increase vs MI350 series), future CPO/NPO ramp with higher stiffener content, and the potential ASIC spec upgrades in the longer term are structural trends for the heat spreader market.

- **Stock implications.** We prefer AVC (OW) in the cooling space given the stronger cold plate spec upgrade trend in both GPU and ASIC systems with its leading position and market share in each project, enabling the company with better-than-peer margin and earnings growth with undemanding valuation (and underperformance vs peers). We also like Fositek (OW) given its near-term GM upside and longer-term QD upgrade/content growth story with continued share gains and TAM expansion.

Investment thesis

We believe the cooling component sector will continue to outperform the market, and in this space, we prefer AVC and Fositek. Compared to other cooling names, these two names 1) still trade at an undemanding valuation (vs historical) with near-term share price performance underperforming other cooling names (or even Taiex for AVC), 2) can continue to deliver both near-term and longer-term earnings upside (e.g. 3Q margin beat, 2028/28E EPS upside from more meaningful spec upgrades), and 3) serve as the leading players in the specific components, enabling the major beneficiaries of spec upgrades and ongoing margin expansion (with smaller-than-expected ASP pressure).

Table 1: We prefer AVC and Fositek in the liquid cooling space

	AVC (3017 TT, OW)	Fositek (6805 TT, OW)	Jentech (3653 TT, OW)
Main cooling product	Cold plate (& Manifold)	Quick disconnect (QD)	Heat spreader/Lid
Main liquid cooling projects/customers	Nvidia VR, major CSP ASICs (Trn3, TPUv8, etc), AMD MI-series	Nvidia VR, major CSP ASICs (Trn3, etc), AMD MI-series	All AI chips (GPU, ASIC, CPU, networking, etc)
JPMc 2026/27/28 EPS	NT\$108/187/236	NT\$64/121/187	NT\$69/150/253
vs BBGe consensus	3%/16%/16% above	1%/12%/40% above	1%/3%/6% above
Share price performance in the past 1 month	Underperforming Taiex by ~2%	Outperforming Taiex by 4%	Outperforming Taiex by 16%
Current P/E valuation (vs historical trading range)	~20x 12M fwd P/E (vs historical 18-22x)	20-25x 12M fwd P/E (vs historical 15-30x)	~50x 12M fwd P/E (vs historical 30-50x)
What has been in the price?	1) AWS Trn3 double-sided liquid cooling design (US\$3-4k per tray content); 2) Nvidia VRU similar or little upgrade vs VR; 3) Google TPU also high content with gold-plated cp design with ~30% market share	1) Some spec upgrades in Nvidia VRU (entry plug QDs/rack manifold socket QDs are more known); 2) new QD customer win (besides AVC)	Both Rubin + Rubin Ultra GPU adopting removable lid as mainstream from 1Q27, with 5-6x content vs 1-piece lid
What has not been fully priced in?	Upside: 1) Nvidia VRU with more meaningful cold plate upgrades, given double-sided design for GPUs + material upgrade to "diamond copper", with ~50% content growth per tray; 2) AMD MI450 already set to adopt double-sided design with higher content; 3) Google TPU v9 also discussing with further upgrades (e.g. double-sided cp) given larger chip size and limited space; 4) more CPO/NPO or optical module adopting cold plates	Upside: 1) more high-spec QD (ZQD) adoption in VRU racks, incl. inside compute tray (for Bianca cp); 2) additional miniQD adoption in VRU for backside cp; 3) more meaningful GM upside in 3Q26; 4) TAM expansion to higher-spec rack manifold QDs from VR200; 5) more QD customer/project wins with stronger share gain potential	Risk: 1) partial Rubin GPU adoption of removable lid (trial volume), 2) only 4-5x content growth vs 1-piece lid when entering MP

Source: Company data, Bloomberg Finance L.P., J.P. Morgan estimates

Nvidia GPU server: VR Ultra likely to see broad-based cooling spec upgrades

Table 2: Nvidia VR Ultra (VRU) system to see meaningful cooling spec upgrades

	GB300 NVL72	Vera Rubin NVL72	Vera Rubin Ultra (VRU) NVL72
Mass production timing	4Q25	3Q/4Q26	Potentially 2H27
Compute tray level:			
Number of compute trays per rack	18	18	18
Number of GPUs per tray	4	4	4
TDP of GPU	1,400W	2,300W	2,600W?
Cooling content per compute tray	~US\$1,500	US\$2,000-2,500	US\$3,000-3,500
- Cold plate (CP)	2 CPs for 2 Bianca	5 CPs for 2 Bianca + 2 CX-9 + 1 PDB	Double-sided design of 6 CPs (2 up + 4 back) for 2 Bianca + 2 CPs for CX-9 + 1 CP for PDB
- Quick disconnect (QD)	10 (8+2)*MQD/UQD	22 (20+2)*MQD/UQD	8*ZQD06 (for 2 Bianca) + 12*MQD + 8*miniQD + 2*ZQD14 (out of tray)
Switch tray level:			
Number of switch trays per rack	9	9	9 (1.5U; incl. 2 layers of board)
Cooling content per switch tray	US\$1,000-1,500	US\$1,500-2,000	>US\$3,000 (incl. 2 boards)
- Cold plate (CP)	2 CPs for each switch IC	1 big CP module on the switch board	2 CP modules on 2 layers of switch board
- Quick disconnect (QD)	20 (18+2)*MQD/UQD	6 (4+2)*MQD/UQD	8*MQD + 2*ZQD14 (out of tray)
Rack level:			
Cooling content summary			
- Compute tray * 18 units	~US\$27,000	~US\$40,500	~US\$58,500
- Switch tray * 9 or 18 units	~US\$11,250	~US\$15,750	~US\$30,000
- Rack manifold * 1 set	~US\$10,000	~US\$10,000	~US\$10,000
Total cooling content per rack (not incl. CDU)	~US\$48,250	~US\$66,250	~US\$98,500

Note: For QD units, the (N+2) means N units of QD inside tray + 2 units of QD (plug QD) outside of tray, so totalling (N+2) units of QD per tray.

Source: J.P. Morgan estimates, Company data.

Compute tray level: cooling content to increase another 30-50% in VRU driven by double-sided cold plate adding diamond copper materials and ZQD adoption

- Cold plate:** We believe the cooling content (including cold plate, QD and inner manifold) of VRU compute tray can reach US\$3,000-3,500 level, which suggests 30-50% content increase vs current VR design, and should be better than market expectation of limited changes or even de-spec. The stronger content growth in VRU compute tray is mainly driven by 1) the **double-sided cold plate design for Bianca modules given VPD (vertical power delivery) adoption (placed at the backside of chip)**: making it 6 cold plates (2 existing + 4 at the backside for rack GPU) on the upper half of tray for the two Bianca modules (vs 2 CP units in VR upper-half tray) + the existing 3 cold plates at the bottom half for the CX-9 chips and power distribution board (PDB), 2) **diamond copper material adoption**: specifically added for the cold plate areas attaching on HBMs, 3) **additional QDs required** (additional 4 sets/8 units of Mini QD, in our view) for the additional cold plates (the ones for the VPD placed at the backside of chip), and 4) **QDs for the Bianca module cold plates upgrading to ZQD** from MQD previously with ~2x ASP increase.
- QD:** In VRU system, we believe Nvidia will upgrade some of its QDs from the previous MQD (Multi Quick Disconnect)/UQD (Universal Quick Disconnect) to the higher-spec ZQD (Zero-drip/flow rate Quick Disconnect) to help improve the cooling efficiency with better liquid flow rate and limited leakage. Hence, we

believe the VRU compute tray QD unit will be largely the same as VR with 20 (inside tray)+2 (outside of tray) units, but the 2 units of plug QD outside of tray will be upgraded to ZQD14 in VRU from UQD08 in previous gens, and 8 of the 20 inside-tray QDs for the two Bianca modules (upper half of tray) will also be upgraded to ZQD06 from MQD06, while the remaining 12 units (for the bottom half of tray) will stay at the MQD06 spec. Aside from the 22 units of MQD+ZQD, we believe there might be an additional 8 units of MiniQD for VRU compute tray, placed at the bottom half of tray. In terms of the QD content per compute tray, we model it to increase to ~US\$750 for VRU vs US\$400-500 for VR.

Switch tray level: 2x QD unit adoption per VRU switch tray (1.5U) with the double-layered (2 boards per tray) architecture

- **Cold plate:** Given the new design of “2 layers of boards per 1.5U switch tray (2 switch boards each at 0.75U)”, we model the cooling content (including cold plate, QD and inner manifold) of the 1.5U switch tray to increase to at least US\$3k, based on 2x cold plate module adopted given 2 switch boards.
- **QD:** The QD units inside the “1.5U” VRU switch tray will likely double vs that in VR, implying an increase to (8+2) units from (4+2) units, given 2 layers/boards per tray. Although the 8 units of QD inside the VRU switch tray should all remain with the MQD06 spec, the 2x unit increase still drives overall QD content growth. In addition, the 2 QDs outside of VRU switch tray will be upgraded to ZQD14 from UQD08 in VR, which also adds incremental content from ASP increase for these 2 plug QDs. In total, we model the QD content per switch tray to increase to ~US\$260 for VRU vs ~US\$150 for VR, based on our estimates.

Table 3: Meaningful QD spec upgrades in VRU with ~100% content increase per NVL72 rack

	GB300 NVL72	Vera Rubin NVL72	Vera Rubin Ultra (VRU) NVL72
Compute tray	18 trays	18 trays	18 trays
- inside-tray QD	8*UQD	20*MQD	12*MQD+8*ZQD06+8*miniQD
- out-of-tray QD	2*UQD (plug)	2*UQD (plug)	2*ZQD14 (plug)
Switch tray	9 trays	9 trays	9 trays (1.5U; incl. 2 layers of board)
- inside-tray QD	18*MQD	4*MQD	8*MQD (given 2 layers/boards)
- out-of-tray QD	2*UQD (plug)	2*UQD (plug)	2*ZQD14 (plug)
Rack manifold (44U)	88*UQD08 (socket)	88*UQD08 (socket)	88*ZQD014 (socket)
Total QD content per NVL72 rack	~US\$9,400	~US\$11,000	~US\$20,500

Source: J.P. Morgan estimates, Company data.

Chip level: 3-piece/removable lid design in Rubin Ultra to drive 4-5x content growth

- **Heat spreader/Lid:** Compared to the current 1-piece lid design in Rubin GPU, we believe Rubin Ultra GPU will adopt the “removable lid + stiffener + multiple screws” design (or so called 3-piece lid design), which should drive the heat

spreader/lid content increase by 4-5x. The Rubin Ultra design will require some screws to attach the lid with the stiffener during chip packaging, and after the packaging and testing process, the screws and lid will then be removed by the ODM vendors to place cold plates on top of the chip with a stiffener surrounding and a TIM1.5 in between.

Table 4: Nvidia GPU heat spreader lid spec/design and content growth

	B300	Rubin	Rubin Ultra
Heat spreader spec/design	1-piece lid	1-piece lid (size increase)	Removable lid +Stiffener + multiple screws
ASP/content per GPU (assuming Rubin ASP=x)	0.6x	x	4-5x
Mass production timing (in terms of heat spreader)	4Q25	2Q26	Potentially 1Q-2Q27

Source: J.P. Morgan estimates, Company data.

CSP ASIC: more aggressive cooling designs and upgrades vs GPU, despite lower TDP

AWS Teton 3 Max (Trainium 3 liquid-cooled system)

Table 5: AWS Trainium 3 (Trn 3) liquid-cooled rack creates higher cooling content vs Nvidia GPU systems

	AWS Trainium 3 NRL 72 (liquid-cooled)	Nvidia Vera Rubin NVL72 (liquid-cooled)
System codename	Teton3 MAX	
Number of racks per universe	2 (NRL72*2 per universe)	N/A
Mass production schedule	3Q/4Q26	3Q/4Q26
Compute tray level:		
Number of compute trays per rack	18	18
Number of ASICs/GPUs per tray	4	4
TDP of ASIC/GPU	700W	2,300W
Cooling content per compute tray	US\$3,000-3,500	US\$2,000-2,500
- Cold plate	Double-sided design of 8 CPs for 4 Trn3 chips + 1 additional tray-sized baseboard CP	5 CPs for 2 Bianca + 2 CX-9 + 1 PDB
- QD	22 units of QD per tray	22 units of QD per tray
Switch tray level:		
Number of switch trays per rack	10	9
Cooling content per switch tray	US\$1,500-2,000	US\$1,500-2,000
- Cold plate	1 cold plate for each switch IC	1 big CP module on the switch board
- QD	6 units of QD per tray	6 units of QD per tray
Rack level:		
Cooling content summary		
- Compute tray * 18 units	~US\$58,500	~US\$40,500
- Switch tray * 10 or 9 units	~US\$17,500	~US\$15,750
- Rack manifold * 1 set	~US\$10,000	~US\$10,000
Total cooling content per rack (not incl. CDU)	~US\$86,000	~US\$66,250

Note: Cooling content includes cold plate, QD, inner manifold and rack manifold (without CDU or sidecar).

Source: J.P. Morgan estimates, Company data.

- **Cold plate:** We believe the cooling content (including cold plate, QD and inner manifold) of AWS Trn3 compute tray can reach US\$3,000-3,500 level (~30% higher than Nvidia's VR compute tray), thanks mainly to the **double-sided cold plate design (8 CPs for 4 ASICs) plus 1 incremental tray-sized cold plate for the baseboard** with decent amount of QD adoption. Also, the Trn3 switch tray may also create US\$1,500-2,000 content value, with slightly more trays per rack (10 switch trays in Teton 3 Max vs 9 units in VR NVL72 rack). Therefore, the **total cooling content (including cold plate, QD and inner manifold) per Teton 3 Max rack could reach ~US\$86k** (vs <US\$70k content for VR NVL72 rack).
- **QD:** AWS Teton 3 Max adopts UQD and MQD in their Trn3 compute and switch tray designs, and the total QD unit adoption per Teton 3 Max rack is **512 units** vs 450 units in Nvidia VR NVL72 rack (both excluding rack manifold socket QDs). Each QD ASP is at ~US\$20, so the total QD content per **AWS Teton 3 Max (Trn3 liquid-cooled rack) is ~US\$10.2k per NRL72 rack**.

- **Heat spreader:** AWS ASICs adopt stiffeners for their heat spreader design, including Trainium 3. The ASP of these general stiffeners ranges between US\$10-20, in our view, but each new generation will still see double-digit ASP growth driven by size increase and material changes.

Google TPU v8 rack (liquid-cooled system)

Table 6: Google TPU v8 compute tray content creates similar cooling content vs Nvidia GPU compute tray despite lower chip TDP

	Google TPU v8 rack (liquid-cooled)	Nvidia Vera Rubin NVL72 (liquid-cooled)
Mass production schedule	3Q/4Q26	3Q/4Q26
Compute tray level:		
Number of compute trays per rack	16	18
Number of ASICs/GPUs per tray	4	4
TDP of ASIC/GPU	~1000W	2,300W
Cooling content per compute tray	US\$2,000-2,500	US\$2,000-2,500
- Cold plate	4 big "gold-plated" CPs for each TPU + 8 small CPs for other surrounding Ics	5 CPs for 2 Bianca + 2 CX-9 + 1 PDB
- QD	2 units of NSP1 QD per tray	22 units of MQD per tray

Note: We only compare compute tray content for TPU rack, as it does not have switch trays (inter-chip interconnect for TPU).

Source: J.P. Morgan estimates, Company data.

- **Cold plate:** We believe the cooling content (including cold plate, QD and inner manifold) of TPU v8 compute tray can reach US\$2-2.5k level, similar to Nvidia's VR compute tray despite lower chip TDP, due to the more complicated cold plate design, including 1) more cold plate unit adoption: a total of 12 cold plates with 4 big ones for TPUs and an additional 8 small ones for surrounding chips, and 2) gold-plating design (given metal TIM adoption) to add incremental content. Therefore, the **total cooling content (including cold plate, QD and inner manifold) per TPU v8 rack could reach ~US\$38k**.
- **QD:** Google TPuv8 compute tray adopts NSP1 QD, unlike Nvidia or other CSP ASIC systems that generally use MQD or UQD, and it only places 2 units of NSP1 QD per TPU compute tray (vs 22 units in VR and Trn3 compute tray). In terms of ASP, we believe NSP1 QD is 2-3x higher than both MQD and UQD. However, based on our calculation, the **total QD content of Google TPU is relatively small at ~US\$1.8k with only 32 units per rack** (vs 400 or 500 unit level for VR and Trn3 rack), despite higher ASP.
- **Heat spreader:** Google TPUs also adopt stiffeners for their heat spreader design, including both v8i and v8t. However, for TPU v9, we believe the removable lid + stiffener could be a potential solution, given much bigger size and higher TDP (potentially reaching 2,000W), which may drive ~10x content increase if the new design is actually adopted. This is yet to be confirmed, but we believe the spec upgrade trend is certain.

Companies Discussed in This Report (all prices in this report as of market close on 29 September 2026, unless otherwise indicated)

Asia Vital Components(3017.TW/NT\$3,400.00/OW), Fositek(6805.TW/NT\$2,515.00/OW)

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Asia Vital Components (3017.TW, 3017 TT) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Aug 26, 2004. All share prices are as of market close on the previous business day.
Break in coverage Feb 27, 2006 - Jan 11, 2024.

Date	Rating	Price (NT\$)	Price Target (NT\$)
11-Jan-24	OW	329.00	420
11-Mar-24	OW	574.00	750
22-Apr-24	OW	630.00	860
10-May-24	OW	654.00	888
20-Jan-25	OW	530.00	820
04-Mar-25	OW	519.00	840
12-Mar-25	OW	560.00	850
05-May-25	OW	507.00	730
03-Jun-25	OW	629.00	820
09-Jul-25	OW	795.00	960
14-Aug-25	OW	1120.00	1,145
14-Sep-25	OW	1050.00	1,200
03-Oct-25	OW	1100.00	1,300
31-Oct-25	OW	1410.00	1,500
13-Nov-25	OW	1505.00	1,700
12-Dec-25	OW	1430.00	1,770
13-Mar-26	OW	1875.00	2,250
25-Mar-26	OW	1945.00	3,000
14-May-26	OW	2585.00	3,200
23-Jul-26	OW	2265.00	3,500
13-Aug-26	OW	2910.00	4,000
29-Sep-26	OW	3555.00	5,200

Fositek (6805.TW, 6805 TT) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Dec 09, 2024. All share prices are as of market close on the previous business day.

Date	Rating	Price (NT\$)	Price Target (NT\$)
10-Dec-24	OW	1020.00	1,320
19-Jan-25	OW	663.00	1,230
04-Feb-25	OW	648.00	1,030
06-May-25	OW	445.00	830
02-Aug-25	OW	916.00	980
14-Aug-25	OW	1045.00	1,170
21-Aug-25	OW	1000.00	1,500
05-Nov-25	OW	1435.00	1,750
12-Nov-25	OW	1560.00	2,000
07-Apr-26	OW	1885.00	2,350
05-May-26	OW	2215.00	2,600
29-Sep-26	OW	2530.00	4,000

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Coverage Universe: Hsueh, Megan : Asia Vital Components (3017.TW), Fositek (6805.TW), Jentech Precision Industrial Co. (3653.TW)

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